



Parshvanath Charitable Trust's
A. P. SHAH INSTITUTE OF TECHNOLOGY
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Department of Computer Science & Engineering (Artificial Intelligence & Machine Learning)

AI-Powered Smart Event Hosting & Management System

- Theme- [SDG Goals](#) (SDG 9 & SDG 11)
- Category- Software
- Group No - 12
- Group Members- 4
- Guide- Prof. Taruna Sharma

Problem Statement:

Event management relies on Google Forms & WhatsApp Manual registration, attendance & feedback analysis. No real-time insights or engagement tracking Inefficient certificate & report generation.

Proposed Solution :

Centralized AI-powered platform for event hosting Automated registration, attendance & feedback. Smart analytics for organizers Single dashboard for students & admins. AI-based feedback sentiment analysis. Smart event recommendations. QR-based attendance with analytics. College-first, low-cost scalable solution.

Innovation & Uniqueness :

AI-based feedback sentiment analysis. Smart event recommendations. QR-based attendance with analytics. College-first, low-cost scalable solution.

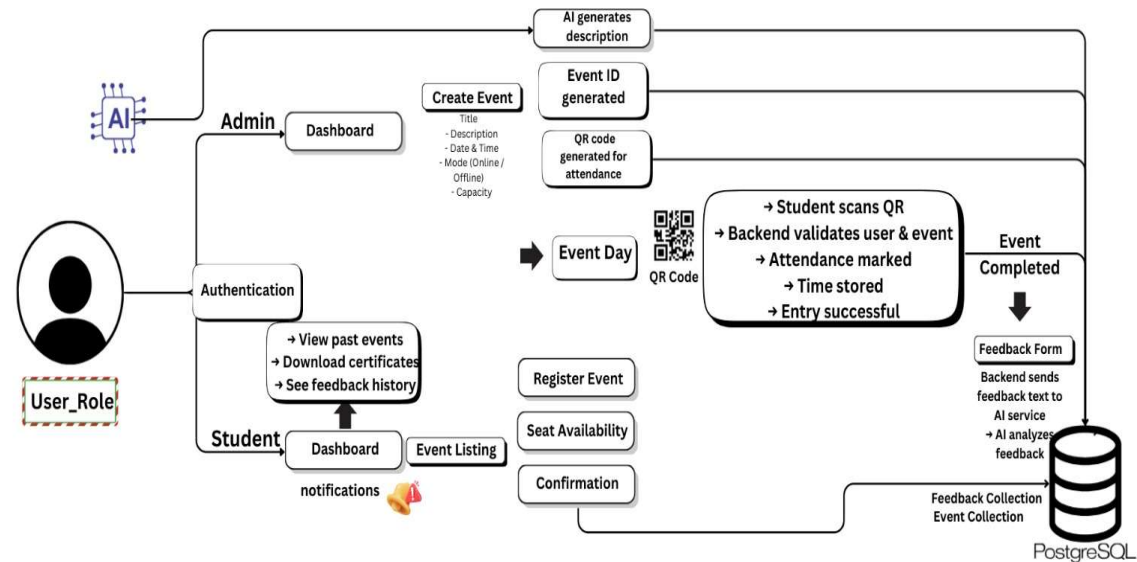
Technical Approach

Technologies Used :

- Frontend: React, Tailwind CSS
- Backend: Node.js, Express
- Database: PostgreSQL
- AI/ML: NLP (Sentiment Analysis), Recommendation Logic
- Authentication: JWT
- Deployment: Vercel, Render, PostgreSQL

Methodology :

- User Authentication & Role Assignment
- Event Creation & Registration
- QR-based Attendance Tracking
- AI Feedback Analysis
- Analytics Dashboard Generation



Feasibility and Viability

Feasibility :

- Uses open-source tools
- No expensive hardware required
- Easily deployable on cloud platforms
- Reusable for multiple semesters

Challenges & Risks :

- Handling large concurrent users
- Data privacy & security
- Accurate AI analysis on small datasets

Mitigation Strategies :

- Cloud scalability
- Secure authentication & access control
- Hybrid rule-based + AI models

Impact and Benefits

Target Audience :

- Colleges & Universities
- Event Organizers
- Students & Participants

Impact :

- Reduces manual workload
- Improves event planning quality
- Data-driven decision making
- Enhances student engagement

Benefits :

- **Social:** Better participation & inclusivity
- **Economic:** Reduces dependency on paid platforms
- **Operational:** Faster & smarter event management

Research and References

References :

- KonfHub – Event Management Platform
- Luma – Event Hosting Platform

Research Papers on:

- AI-based Feedback Analysis
- Recommendation Systems
- Event Engagement Analytics

Research Insight :

- Existing platforms lack academic customization
- AI improves insight extraction & engagement tracking
- Demand for affordable campus-level solutions